## 1.06 Quadratics: Introduction

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# 1: Solving Quadratic Equations

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$$x^2 + 5x = 6$$

There are three ways of solving a quadratic equation.
One is by *completing the square*, which we'll do later.
What are the other two ways?

By factorisation

Put in form.
$ax^2 + bx + c = 0$
with 0 on one side

Factorise. If the product of two things is 0, at least one must be 0.

Using the quadratic formula.

If $ax^2 + bx + c = 0$ then

$x^2 + 5x - 6 = 0$

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# 1: Solving Quadratic Equations

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Solving without factorising

If the subject only appears once however, it might be easier not to expand out/factorise:

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### Test Your Understanding

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1 Solve $(x+3)^2 = x+5$ using factorisation.

2 Solve $(2x+1)^2 = 5$

3 Solve $\sqrt{x+3} = x-3$

4 Solve $2x + \sqrt{x} - 1 = 0$

**"Use of Technology":**
The 'equation' mode on your calculator will solve quadratics in the form $ax^2+bx+c=0$. When you're asked for the 'order', use 2 (we'll see why later).

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### Exercise 2A/2B Pure Y1 Pages 20-22

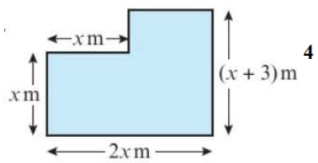
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**3** Solve the following equations:

**i** $6x^2 - 7 = 11x$

**h** $5x^2 - 10x^2 = -7 + x + x^2$

**4** This shape has an area of $44\,\text{m}^2$. Find the value of $x$.




**4**

**5** Solve the equation $5x + 3 = \sqrt{3x+7}$.

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### Exercise 2A/2B Pure Y1 Pages 20-22

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**2** Solve the following equations using the quadratic formula. Give your answers to three significant figures.

**h** $2x^2 + 11x = 5x^2 - 18$

**3** For each of the equations below, choose a suitable method and find all of the solutions. Where necessary, give your answers to three significant figures.

**e** $(2x + 8)^2 = 100$ **f** $6x^2 + 6 = 12x$

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### Exercise 2A/2B Pure Y1 Pages 20-22

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**Extension:** (Full Database: http://www.drfrostmaths.com/resources/resource.php?rid=268 )

1 **(i)** Use the substitution $\sqrt{x} = y$ (where $y \geqslant 0$) to find the real root of the equation

$$x + 3\sqrt{x} - \tfrac{1}{2} = 0\,.$$

**(ii)** Find all real roots of the following equations:

**(a)** $x + 10\sqrt{x+2} - 22 = 0\,;$

**(b)** $x^2 - 4x + \sqrt{2x^2 - 8x - 3} - 9 = 0\,.$

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# 2: Completing the Square

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"Completing the square" means putting a quadratic in the form $(x+a)^2+b$ or $a(x+b)^2+c$

The **underlying reason we do this is because $\mathbf{x}$ only appears once in the expression** (e.g. in $(x+2)^2+3$ vs $x^2+4x+7$), which makes it algebraically easier to handle. This has a number of consequences:

a. Solving Quadratics

If we have a completed square:

$$(x+4)^2-7=0$$

we saw at the start of the chapter how we could rearrange to make $x$ the subject.
Indeed using the quadratic formula is actually solving the quadratic by completing the square – it's just someone has done the work for us already!

b. Sketching Quadratics

We'll see later that if $y=(x+a)^2+b$, then the minimum point is $(-a,b)$

c. In integration

In Further Maths, completing the square allows us to 'integrate' expressions like:

$$\int \frac{1}{x^2-4x+5}\,dx$$

(you will cover integration later this module)

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# Completing the Square Recap

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Expand:
$(x+\mathbf{9})^2=$
$(x-\mathbf{5})^2=$
What do you notice about the relationship between the bold numbers?

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# Completing the Square

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Express $2x^2 + 12x + 7$ in the form $a(x + b)^2 + c$

Factorise out coefficient of $x^2$. You can leave the constant term outside the bracket.

Complete the square inside the bracket (you should have two sets of brackets)

Expand out outer bracket.

Express $5 - 3x^2 + 6x$ in the form $a - b(x + c)^2$

It may help to write in the form $ax^2 + bx + c$ first.

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# Test Your Understanding

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Express $3x^2 - 18x + 4$ in the form $a(x + b)^2 + c$

Express $20x - 5x^2 + 3$ in the form $a - b(x + c)^2$

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### Solving by Completing the Square

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Solve the equation:

$$3x^2 - 18x + 4 = 0$$

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### Exercise 2C/2D Pure Y1 Page 23-24

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**1** Complete the square for the expressions:

**a** $x^2 + 4x$ **b** $x^2 - 6x$ **c** $x^2 - 16x$ **d** $x^2 + x$ **e** $x^2 - 14$

**2** Complete the square for the expressions: **a** $2x^2 + 16x$ **b** $3x^2 - 24x$ **c** $5x^2 + 20x$ **d** $2x^2 - 5x$ **e** $8x - 2x^2$

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# Exercise 2C/2D Pure Y1 Page 23-24

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4 Given that $x^2 + 3x + 6 = (x + a)^2 + b$ find the values of the constants $a$ and $b$

5 Write $2 + 0.8\text{x} - 0.04x^2$ in the form $A - B(x + C)^2$ where $A \text{and} B$ are constants to be determined

**4** By completing the square, show that the solutions to the equation $x^2 + 2bx + c = 0$ are given by the formula $x = -b \pm \sqrt{b^2 - c}$. (4 marks)

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# Proving the Quadratic Formula

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If $ax^2 + bx + c = 0$, prove that $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Just use exactly the same method as you usually would!